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Local planning and evaluation of evasive maneuvers for automated vehicles

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Abstract:

Autonomous vehicles must plan safe evasive trajectories across a range of real-world obstacle configurations. This report documents the design, mathematical formulation, and Python-based implementation of a local trajectory planning system and validate using CARLA. Three progressively complex scenarios are addressed: overtaking a stationary obstacle, overtaking a slower-moving vehicle under uncertain speed conditions, and navigating a pedestrian who crosses the road in front of a parked vehicle.

The trajectory planner employs quintic polynomials to generate geometrically smooth, continuous lane-change profiles evaluated via the kinematic bicycle model. An adaptive counter-steering scaling function modulates curvature overshoot in a physically motivated manner. Trajectories are selected by a weighted cost function balancing passenger comfort, safety margin, maneuver economy, and speed preference. Scenario 3 introduces a prospective pedestrian safety check, the planner predicts whether the complete future trajectory stays outside the exclusion zone before permitting the vehicle to resume overtaking after braking and waiting.

All scenarios were validated through Matplotlib visualization and CARLA autonomous driving simulator. Scenario 1 produced a maneuver time of 13.03 s with maximum lateral acceleration 1.09 m/s^2 . Scenario 2 achieved overtaking at 18.0 km/h against a 10.1 km/h obstacle with minimum clearance 1.95 m. Scenario 3 correctly triggered BRAKE_WAIT_UNTIL_SAFE_THEN_OVERTAKE mode, achieving a final pedestrian radius margin of 3.555 m despite the raw margin at the original decision point being 1.031m.

Keywords: *Trajectory Planning, Automated Driving, Quintic Smoothstep, Lane Change, Kinematic Bicycle Model, Pedestrian Safety, CARLA Simulator, Evasive Maneuver, Grid Search Optimization, Python*

1. Introduction:

1.1 Background:

Autonomous driving has transitioned from research prototype to commercial deployment over the past decade. At the heart of every autonomous driving stack lies the motion planning subsystem, which translates perceived environmental state and high-level mission goals into dynamically realizable vehicle trajectories. Among the planning layers, local (or trajectory) planning is the most computationally intensive: it must generate geometrically precise paths, enforce kinematic and dynamic constraints in real time, guarantee minimum safety margins against all detected obstacles, and produce smooth motion profiles that are comfortable for passengers and survivable for actuators [1].

This project develops and validates a local planner capable of executing evasive maneuvers across a progression of driving scenarios. The project's states secondary objectives include Python implementation, CARLA simulation testing, and preparation of the ANTON vehicle platform for eventual real-world deployment.

1.2 Problem Statement:

The engineering problem is: given a two-lane straight road with an ego vehicle and one or more obstacles (a stationary vehicle, a slower-moving vehicle, or a stationary vehicle combined with a crossing pedestrian), generate safe, smooth, kinematically feasible overtaking trajectories or appropriate fallback behaviors that satisfy all hard constraint bounds and minimize a cost function encoding comfort, safety, and efficiency.

1.3 Objectives:

1. Design a continuous path generator using quintic polynomials.
2. Implement the kinematic bicycle model for constraint derivation.
3. Implement adaptive counter-steering scaling to suppress lateral acceleration overshoot.
4. Develop an optimization-based trajectory planning (Optimal Control / MPC-like approach) with cost-function ranking for all three scenarios.
5. Implement scenario-specific safety: inflated bounding-box clearance and pedestrian radius margin.
6. Validate through CARLA synchronous-mode replay.

2. Literature Review:

2.1 Overview of Local Path Planning Categories:

Local trajectory planning methods for automated driving fall into three principal categories: sampling-based methods, curve interpolation methods, and machine learning/deep learning methods [2]. Each offers a distinct tradeoff between computational cost, solution quality, safety guarantees, and implementation complexity [3]. The following subsections summarize each category and presents a structured comparison that justifies the selected approach.

2.2 Sampling-Based Methods:

Rapidly Exploring Random Trees (RRT) and its optimal variant RRT* explore configuration space through stochastic sampling. RRT builds a space-filling tree by drawing random samples and extending the nearest tree node toward each sample; RRT* additionally rewires the tree to improve cost. Both algorithms are probabilistically complete and handle arbitrary obstacle shapes, but they produce non-smooth paths requiring post-processing and offer no deterministic runtime guarantees [4]. Hybrid A*, developed for the DARPA Urban Challenge, combines grid search with forward-simulated kinematic vehicle models, producing drivable paths even in structured environments but its curvature can be discontinuous without explicit smoothing, and the search space grows steeply with resolution.

2.3 Curve Interpolation Methods:

Curve interpolation methods fit parametric curves through boundary-condition waypoints derived from the initial and desired final states. Polynomial functions of order n are the dominant choice. It demonstrates safe and optimal lane-change paths using fifth-order polynomials that enforce position, velocity, and acceleration matching at both endpoints. Trigonometric functions (sinusoidal profiles) are also used, offering analytical tractability but without natural boundary enforcement [5]. The Frenet-frame planner of Werling [6] systematically applies quintic polynomials in a road-aligned coordinate frame, sampling families of end-state conditions and selecting the minimum-cost candidate, a framework directly related to the approach adopted in this project.

2.4 Machine Learning Deep Learning Methods:

Data-driven approaches train models on expert demonstrations or interaction logs. Support Vector Machines and shallow neural networks have been used for lane-change prediction [7]. Deep architectures including Long Short-Term Memory networks and deep belief networks learn spatiotemporal driving patterns [8]. These methods can capture nuanced multi-agent interactions and do not require explicit constraint formulations, but they require large, high-quality training datasets, offer limited interpretability, do not easily provide hard safety guarantees, and generalize poorly to configurations outside the training distribution. For a project with explicit constraint verification requirements and a limited dataset, this category is unsuitable.

3. Method Comparison and Justification:

3.1 Quantitative Comparison of Planning Methods:

Table 1 provides a structured comparison of the most relevant planning approaches against the nine criteria that matter most for this project. The quintic polynomial (MPC) approach, our selected method, is shown in the highlighted green column. Ratings use a 1–5 scale (5 = best) and are justified in the discussion below the table.

Table 1: Comparison of different planning methods

Criterion	RRT / RRT*	Hybrid A*	Trigonometric	Quintic Polynomial (MPC)
Path Continuity (zero boundary curvature)	2 post-processing required	3 Approx. only	3 Enforceable with constraints	5 Analytic guarantees
Deterministic Runtime	1 Probabilistic, unbounded	3 Bounded but resolution-dependent	5 Closed forms	4 Linear in grid size
Parameter Interpretability	1 Abstract graph nodes	2 Heuristic weights	3 Amplitude, wavelength	4 End-state boundary values
Safety Guarantee (hard constraint verification)	3 Probabilistic	4 Deterministic on discrete grid	4 Verifiable	5 Exact per-step check
Comfort (lateral accel. smoothness)	2 Jerky without smoothing	3 Moderate	4 Good	5 Excellent
Computational Cost (offline, prototyping)	1 High (iterative tree)	3 Medium (grid search)	5 Very low	4 Low (grid search)
Multi-Speed / Multi-Agent Extensibility	4 Natural	3 Moderate	2 Fixed speed assumption	4 Frenet frame handles curvature
Pedestrian Interaction Support	3 Possible with replanning	3 Possible with replanning	2 Not natural	4 Decision point insertable
Real-World Deployment Readiness (current scope)	2 Needs closed-loop controller	3 Needs controller	3 Needs controller	4 Needs controller

Notes on scoring:

- Continuity is a hard requirement for our project: any method that requires post-processing to achieve it (RRT) or can only approximately enforce it (Hybrid A*, trigonometric) introduces risk that a boundary curvature spike will violate the ay constraint. Quintic polynomials guarantee analytically, no post-processing needed.

- Deterministic runtime is critical for a real-time planning system and for unit-testing reproducibility. RRT's probabilistic nature means the same problem instance can return very different solutions or fail to converge. The closed-form quintic evaluation with a bounded grid guarantees the same result every run.
- Parameter interpretability matters both for engineering validation and for the academic setting: a supervisor must be able to read the parameter table (Table B.3) and understand physically what each parameter does. Saying 'outward lane-change length = 12 m' is immediately meaningful. Saying 'RRT sample budget = 5000 nodes' is not.
- The pedestrian interaction support distinguishes our method: no other approach in the table directly supports the brake-wait-continue logic without a full replanning event. Our method builds this logic explicitly on top of the planned trajectory, making the safety verification transparent and auditable.

3.2 Why Quintic Polynomial (Optimal Control / MPC-like approach)?

Frenet-frame polynomial planning [6] and our quintic smoothstep approach share the same polynomial class and the same C2 guarantees. The key differences that led us to choose the smoothstep formulation are:

Table 2: Frenet-frame Polynomial VS quintic Polynomial

Aspect	Frenet Frame Polynomial	Quintic Polynomial (Our Method)
Coordinate frame	Road-aligned Frenet (s, d); requires road centerline extraction	Global Cartesian; no road geometry preprocessing needed
End-state sampling	Samples arbitrary (d _f , t _f) pairs; large combinatorial space	Physical parameters (L _{out} , gap) directly constrain the grid
Boundary conditions	Polynomial coefficients fitted to sampled end states	Coefficients fixed by smoothstep formula; no fitting step
Lateral profile shape	Varies with sampled end state; can over- or undershoot	Strictly sigmoidal; guaranteed to monotonically traverse the lane
Counter-steering	Not inherently modelled; emerges from polynomial derivative	Explicitly scaled by adaptive α function for comfort control
Pedestrian logic	Not described in original paper; requires extension	Built natively into Scenario 3 brake-wait-continue architecture
Code complexity	Moderate (coordinate transforms, polynomial fitting)	Low (closed-form evaluation, no fitting)

In summary, the quintic smoothstep polynomial approach delivers the same mathematical guarantees as Frenet polynomial planning while removing the need for road geometry preprocessing, providing physically interpretable grid parameters, and naturally accommodating

the pedestrian interaction logic required by Scenario 3. For a straight-road overtaking problem implemented in Python without a full HD-map pipeline, this is the optimal choice among the methods considered.

4. System Requirements and Architecture:

4.1 Functional Requirements:

Table 3: Functional Requirements

ID	Description
FR-01	Generate lane-change trajectories parameterized by outward length, return length, start gap, and return gap.
FR-02	Enforce minimum 0.5 m clearance between ego and obstacle inflated bounding boxes at all time steps.
FR-03	Reject candidates exceeding maximum lateral acceleration (3.5 m/s ² Scenario 1; 3.8 m/s ² Scenarios 2–3).
FR-04	Reject candidates exceeding maximum steering angle $\pm 25^\circ$.
FR-05	Rank all feasible candidates by cost function; select minimum.
FR-06	Compute safe braking fallback if no feasible overtake exists (Scenario 2).
FR-07	Detect pedestrian conflicts; generate brake-wait-continue trajectory (Scenario 3).
FR-08	Output time-stamped arrays (x, y, ψ , δ , ay, t, obs_x) for CARLA replay.

4.2 Non-Functional Requirements:

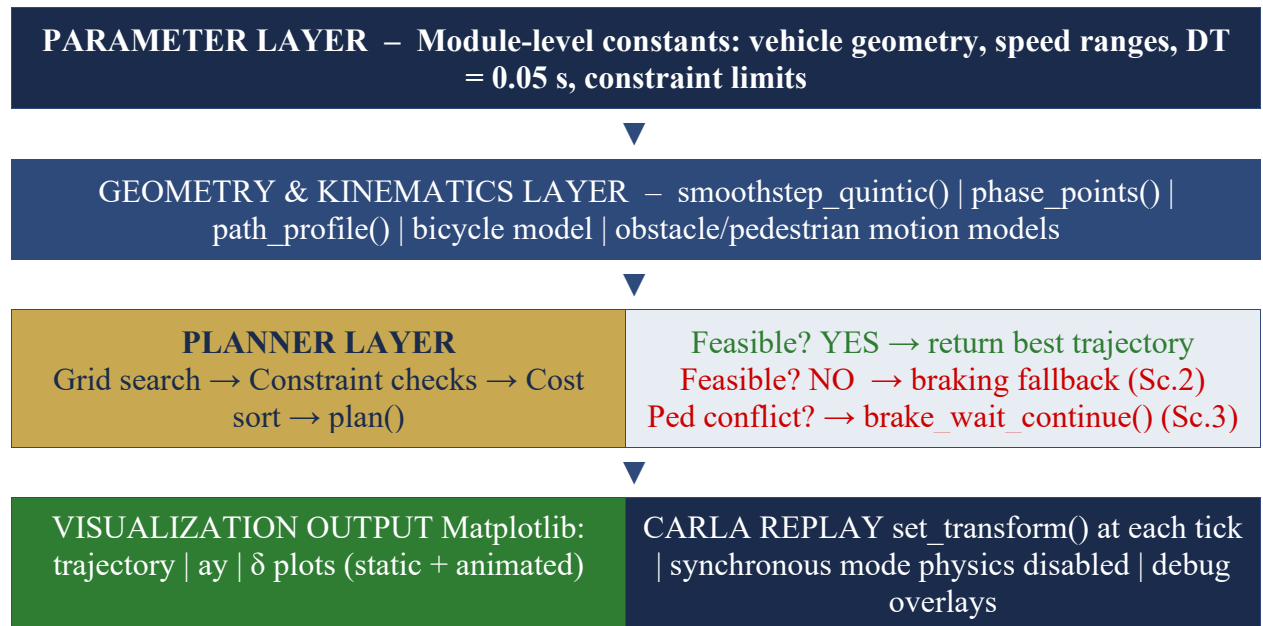
Table 4: Non-Functional Requirements

ID	Description
NFR-01	Trajectory boundary conditions shall be continuous (zero lateral velocity and curvature at transitions).
NFR-02	Planning shall complete within a few seconds on a standard laptop for all scenarios.
NFR-03	System shall be modular: generator, constraint checker, cost evaluator, visualization separated.
NFR-04	All physical parameters shall be named module-level constants.
NFR-05	CARLA replay shall operate in synchronous mode at fixed DT = 0.05 s.

4.3 System Architecture Flowchart:

Figure 1 illustrates the four-layer pipeline and the data dependencies between modules.

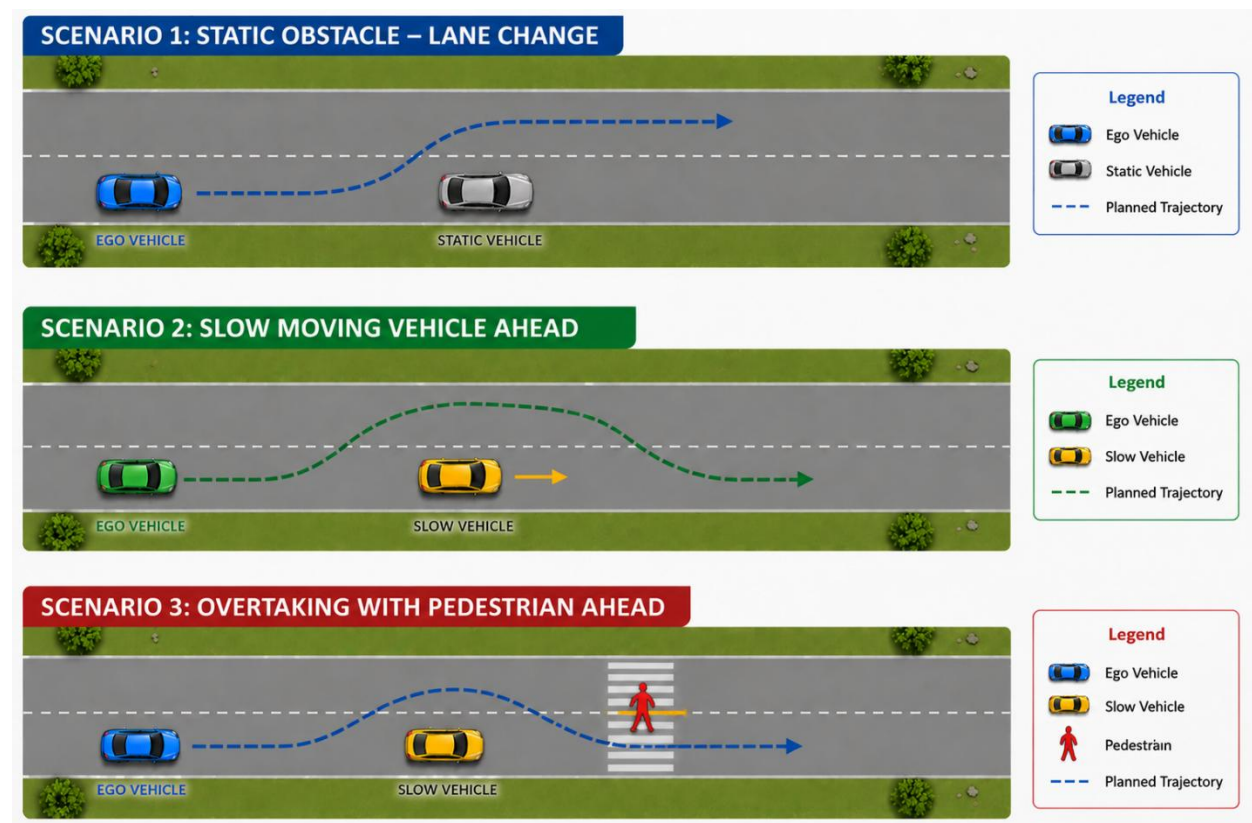
Figure 1: Four-layer pipeline and the data dependencies



5. Scenario Definitions:

The three scenarios are shown in Figure 2 below. Each build on the previous by adding a new agent or interaction mode.

Figure 2: Different scenarios for ego vehicle



5.1 Scenario 1:

5.1.1 Static Obstacle - Lane Change:

A stationary vehicle is at $x_{\text{obs}} = 50.0\text{m}$ in the ego lane ($y = 0$). The ego approaches and executes a smooth left-side lane change, passes in the adjacent lane at $y = 3.75\text{m}$, and returns to the original lane. Phase points are computed from bumper gap parameters alone, since the obstacle does not move.

5.2 Scenario 2:

5.2.1 Slow-Moving Vehicle Ahead:

A slower vehicle moves at $v_{\text{obs}} \in [2.8, 3.8]$ m/s starting at $x_{\text{obs}} = 20.0\text{m}$. The ego searches $v_{\text{ego}} \in [4.0, 5.0]$ m/s with a minimum relative speed of 1.0 m/s. Phase points are derived using the Doppler scaling factor $(1 - v_{\text{obs}}/v_{\text{ego}})$. If no feasible overtake is found, a safe braking trajectory is computed.

5.3 Scenario 3:

5.3.1 Overtaking with Pedestrian Crossing:

A stationary vehicle is at $x_{\text{obs}} = 30.0\text{m}$. A pedestrian crosses the road at $x_{\text{ped}} = 38.0\text{m}$, starting at $y = -3.0\text{m}$ and moving at 1.0 m/s toward $y = 10.0\text{m}$. The pedestrian safety radius is $2.5\text{m} + 0.25\text{m}$ buffer. The planner first selects the best overtaking trajectory then checks whether the future ego body path stays outside the exclusion zone. If not, it activates the brake-wait-continue logic.

6. Mathematical Foundations:

6.1 Coordinate System:

Local Cartesian frame: x forward, y left (positive = overtaking lane), z up. Heading angle ψ counterclockwise from x-axis. CARLA world conversion:

$$\begin{aligned} \text{world_x} &= \text{base_x} + x \cdot \cos(\text{base_yaw}) - y \cdot \sin(\text{base_yaw}) \\ \text{world_y} &= \text{base_y} + x \cdot \sin(\text{base_yaw}) + y \cdot \cos(\text{base_yaw}) \end{aligned}$$

6.2 Quintic Polynomial:

The lateral displacement for each lane-change phase uses a normalized progress variable $u \in [0, 1]$:

$$S_5(u) = 10u^3 - 15u^4 + 6u^5$$

This satisfies $S_5(0) = 0$, $S_5(1) = 1$, $S_5'(0) = S_5'(1) = 0$, $S_5''(0) = S_5''(1) = 0$, guaranteeing C2 continuity at all phase boundaries. Derivatives:

$$S_5'(u) = 30u^2 - 60u^3 + 30u^4; \quad S_5''(u) = 60u - 180u^2 + 120u^3$$

For the outward lane-change on $[x_1, x_2]$ with $u = (x - x_1) / L_{\text{out}}$:

$$y = y_t \cdot S_5(u); \quad dy/dx = y_t \cdot S_5'(u) / L_{\text{out}}; \quad d^2y/dx^2 = y_t \cdot S_5''(u) / L_{\text{out}}^2$$

where $y_t = 3.75\text{m}$ (lane width). The passing phase $[x_2, x_3]$ is straight at $y = y_t$. The return phase $[x_3, x_4]$ uses $1 - S_5(u)$ with sign-reversed derivatives.

6.3 Adaptive Counter-Steering Scaling:

The counter-curvature component (negative d^2y/dx^2 in outward phase; positive in return phase) is scaled by $\alpha \in [0.40, 0.85]$:

$$\alpha = \text{clip}(0.45 + 0.20 \cdot f_L + 0.15 \cdot f_g - 0.10 \cdot f_v, 0.40, 0.85)$$

where $f_L = (L_{lc} - 12)/8$ rewards longer maneuvers (more damping room), $f_g = (g - g_{ref})/g_{range}$ rewards larger initial gaps, and $f_v = v_{rel}/v_{scale}$ penalizes high relative speeds. The Scenario 1 speed coefficient is -0.08 (slightly less penalty) vs -0.10 for Scenarios 2–3.

6.4 Kinematic Bicycle Model:

For wheelbase $L = 1.686$ m, heading angle ψ , path curvature κ , steering angle δ , and lateral acceleration a_y :

$$\psi = \arctan(dy/dx); \quad \kappa = (d^2y/dx^2) / (1 + (dy/dx)^2)^{3/2}; \quad \delta = \arctan(L \cdot \kappa); \quad a_y = v^2 \cdot \kappa$$

6.5 Phase Points for Moving Obstacles:

$$x1 = (x_{obs0} - (L_{obs} + L_{ego})/2 - g_{start}) / (1 - v_{obs}/v_{ego})$$

$$x4 = (x_{obs0} + (L_{obs} + L_{ego})/2 + g_{return}) / (1 - v_{obs}/v_{ego})$$

The Doppler scaling factor $(1 - v_{obs}/v_{ego})$ approaches zero as $v_{obs} \rightarrow v_{ego}$, correctly pushing $x1, x4 \rightarrow \infty$ when overtaking becomes geometrically impossible.

6.6 Inflated Bounding-Box Clearance:

$$dx = |x_{ego} - x_{obs}| - (L_{obs} + L_{ego})/2; \quad dy = |y_{ego} - y_{obs}| - (W_{obs} + W_{ego})/2$$

$$clearance = \sqrt{\max(dx, 0)^2 + \max(dy, 0)^2} + \min(\max(dx, dy), 0)$$

Positive outside the inflated box, zero on boundary, negative inside. Minimum over all time steps must exceed 0.5m.

6.7 Pedestrian Safety (Scenario 3):

$$dx_b = \max(|x_{ego} - x_{ped}| - L_{ego}/2, 0); \quad dy_b = \max(|y_{ego} - y_{ped}| - W_{ego}/2, 0)$$

$$margin = \sqrt{dx_b^2 + dy_b^2} - (r_{safe} + b) = \sqrt{dx_b^2 + dy_b^2} - 2.75$$

$margin \geq 0$ must hold for every future time step from the decision point onward before the vehicle may resume overtaking.

6.8 Cost Function:

$$J = 0.3 \cdot \text{mean}(a_y^2) + 0.05 \cdot L_{maneuver} + 25/(clearance + 10^{-3}) + 0.03 \cdot \text{gap_error} - 0.2 \cdot v_{ego}$$

Table 5: Cost function terms definition

Term	Rationale
$0.3 \cdot \text{mean}(a_y^2)$	RMS lateral acceleration – ISO 2631 comfort proxy.
$0.05 \cdot L_{maneuver}$	Penalizes unnecessarily long maneuvers (Scenarios 2–3).
$25/(clearance + 10^{-3})$	Inverse barrier – strongly penalizes near-minimum clearances.
$0.03 \cdot \text{gap_error}$	Deviation from desired bumper gap encourages consistent gaps.
$-0.2 \cdot v_{ego}$	Rewards higher ego speed – throughput vs. comfort tradeoff.

7. Implementation:

7.1 Development Environment:

Table 6: Environment definition

Component	Detail
Language	Python 3.10+
Numerical	NumPy (vectorized array operations, linspace, gradient, arctan, sqrt)
Visualization	Matplotlib 3.x (static plots; FuncAnimation for animated replay)
Simulation	CARLA 0.9.x (synchronous mode, fixed_delta_seconds = 0.05 s)
CARLA API	carla Python package (actor spawning, transform, debug draw)
Pygame	v2.x (multi-camera window for Scenario 2 replay)
IDE	VS Code; tested on Windows 10/11 and Ubuntu 22.04

7.2 Algorithm Workflow – Pseudocode:

PROCEDURE plan():

INIT grid: $lc_out_lengths \times lc_return_lengths \times start_gaps \times return_gaps \times speeds$

FOR each (v_ego , v_obs , L_out , L_ret , g_start , g_ret):

IF $v_ego - v_obs < RELATIVE_SPEED_MIN$: CONTINUE

Compute phase points ($x1, x2, x3, x4, L_pass$) $\leftarrow$ Doppler scaling for moving obstacle

IF $L_pass \leq 0$: CONTINUE

Generate x array; $path_profile() \rightarrow y, dy/dx, d^2y/dx^2 \leftarrow$ quintic smoothstep

Bicycle model $\rightarrow \psi, \kappa, \delta, ay$; compute $obs_x(t), [ped_y(t)]$

IF $\max|ay| > ay_max$: CONTINUE

IF $\max|\delta| > delta_max$: CONTINUE

IF $\min_clearance(ego, obstacle) < d_min$: CONTINUE

[Sc.3] Compute future pedestrian body margin from $x2$ onward

Compute cost J; append to $feasible_results$

SORT $feasible_results$ by J; SELECT $feasible_results[0]$

[Sc.3] IF selected trajectory violates ped zone $\rightarrow$ brake_wait_continue()

[Sc.2] IF $feasible_results$ empty $\rightarrow$ braking_fallback()

7.3 Key Function Summary:

Table 7: Key Function Summary

Function	Description
smoothstep_quintic() + derivatives	Closed-form evaluation of $S_s(u)$, $S_s'(u)$, $S_s''(u)$. Analytical; avoids discretization error in κ .
phase_points()	Computes $(x_1-x_4, L_{\text{pass}})$ with Doppler scaling. $L_{\text{pass}} \leq 0$ triggers early rejection.
path_profile()	Evaluates y , dy/dx , scaled d^2y/dx^2 at all trajectory samples across three phases.
signed_distance_to_inflated_obstacle()	Per-step signed clearance using inflated bounding box. Min value is the feasibility metric.
ego_rectangle_to_pedestrian_clearance()	Distance from pedestrian centre to nearest point on ego body rectangle. More conservative than point-point.
find_safe_brake_start_index() [Sc.3]	Scans backward from decision index until ego body stays outside ped radius throughout brake+wait.
find_safe_continue_time() [Sc.3]	Scans forward from stop time until full future trajectory maintains non-negative ped margin.
create_brake_after_lane_change_result()	Concatenates pre-brake braking wait re-acceleration segments into unified time-synchronized dict.

8. Results and Evaluation:

8.1 Scenario 1:

8.1.1 Static Obstacle Overtaking:

The scenario 1 planner enumerated 2704 candidate trajectories. The terminal output showing the exact maneuver timing and counter-steering breakdown is reproduced in Figure 3.

Figure 3: Scenario 1 trajectory plot – all feasible trajectories (blue) and selected optimal (red)

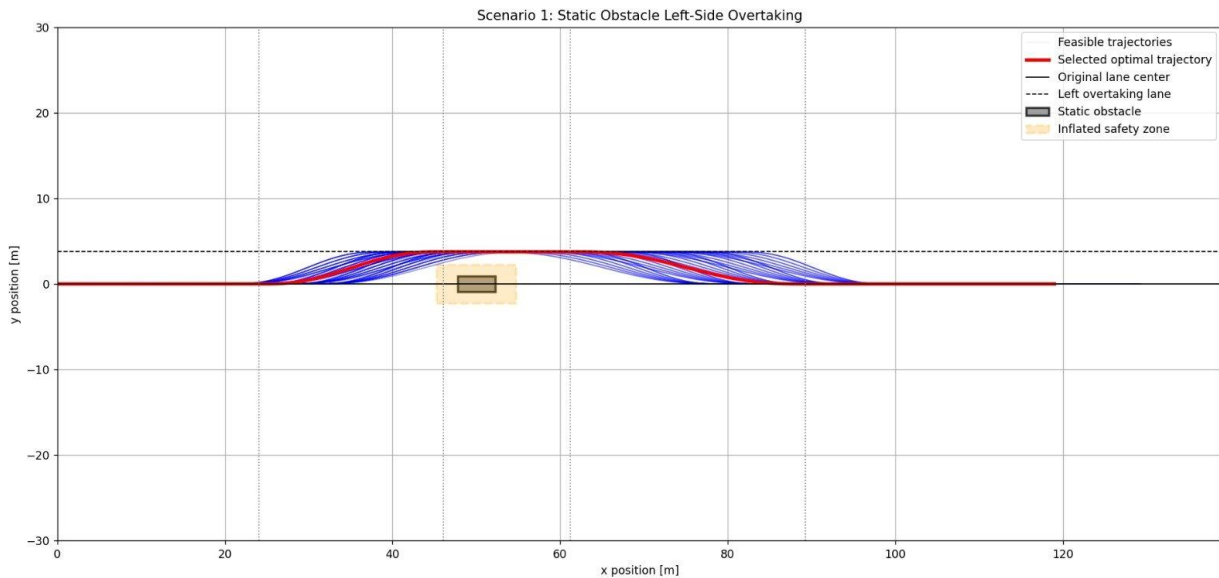


Table 8: Scenario 1 terminal output

Metric	Value	Limit / Target	Status
Feasible trajectories	2704	—	—
Ego speed	18.0 km/h	—	—
Outward lane-change length	22.0 m	—	—
Return lane-change length	28.0 m	—	—
Passing length	15.17 m	—	—
Cost value	11.9037	—	—
Max lateral acceleration	1.0876 m/s ²	—	—
Maximum steering angle	4.1952°	—	—
Outward main a_y	1.0876 m/s ²	$\leq 3.5 \text{ m/s}^2$	✓ PASS
Outward counter a_y	0.7856 m/s ²	$< \text{main } a_y$	✓ PASS
Return main a_y	0.6783 m/s ²	$\leq 3.5 \text{ m/s}^2$	✓ PASS
Return counter a_y	0.5765 m/s ²	$< \text{main } a_y$	✓ PASS
Outward counter scale α	0.722	[0.40, 0.85]	✓ PASS
Return counter scale α	0.850	[0.40, 0.85]	✓ PASS
Lane-change time	~4.4 s	—	—

Passing time	3.033 s	—	—
Return time	5.6 s	—	—
Total maneuver time	13.033 s	—	—
Overtaking side	LEFT (positive y)	LEFT only	✓ PASS

Figure 3 shows the overhead trajectory plot. The fan of blue semi-transparent lines represents all feasible candidate trajectories; the selected optimal trajectory is shown in red. The gray rectangle is the obstacle bounding box; the orange zone is the inflated 0.5 m safety boundary.

The lateral acceleration profile (Figure 4) shows the characteristic asymmetric main-pulse / counter-pulse pattern from the quintic polynomial. Both phases remain well below 3.5 m/s². The steering angle profile (Figure 5) mirrors this, peaking near $\pm 5^\circ$ with zero crossings at each phase boundary — confirming the C2 condition analytically.

Figure 4: Scenario 1 lateral acceleration profile – main and counter-steering pulses within limits

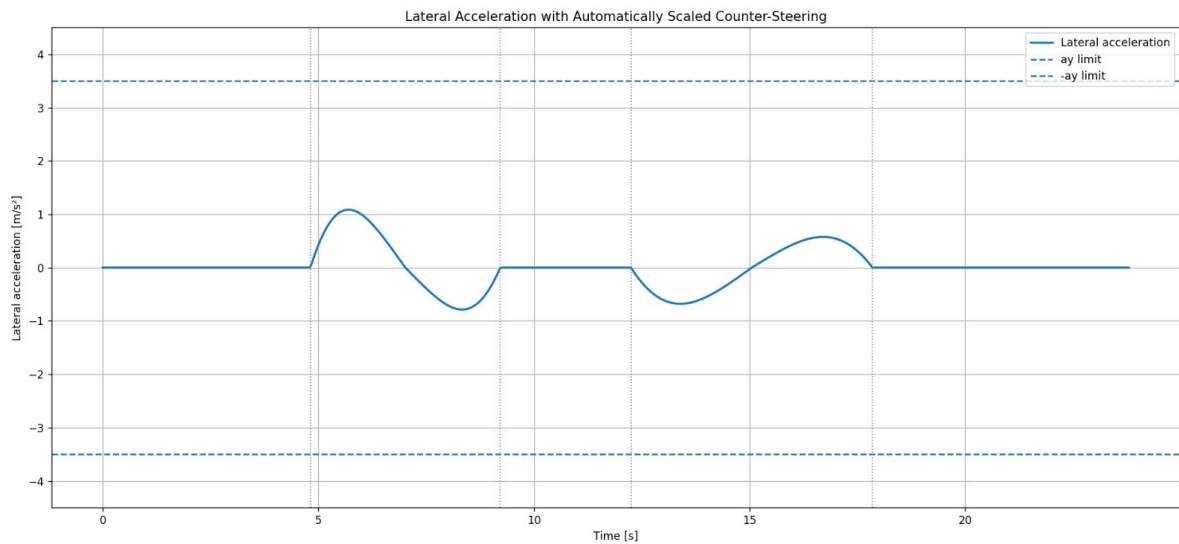
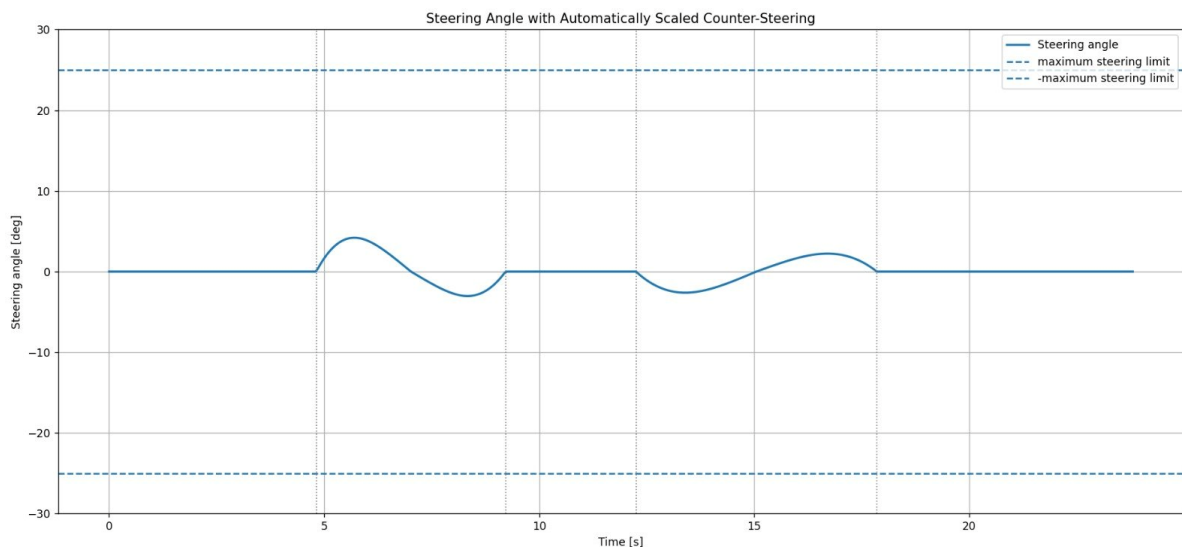


Figure 5: Scenario 1 steering angle profile – well within the $\pm 25^\circ$ constraint



8.2 Scenario 2:

8.2.1 Slow-Moving Vehicle Ahead:

Scenario 2 extended the search space to include joint ego and obstacle speed dimensions. The animated Matplotlib snapshot (Figure 6) captures the maneuver at $t = 10.80$ s with the ego in the passing lane alongside the moving obstacle. The terminal results are in table 9.

Figure 6: Scenario 2 animated trajectory snapshot – ego (red) completing the passing phase

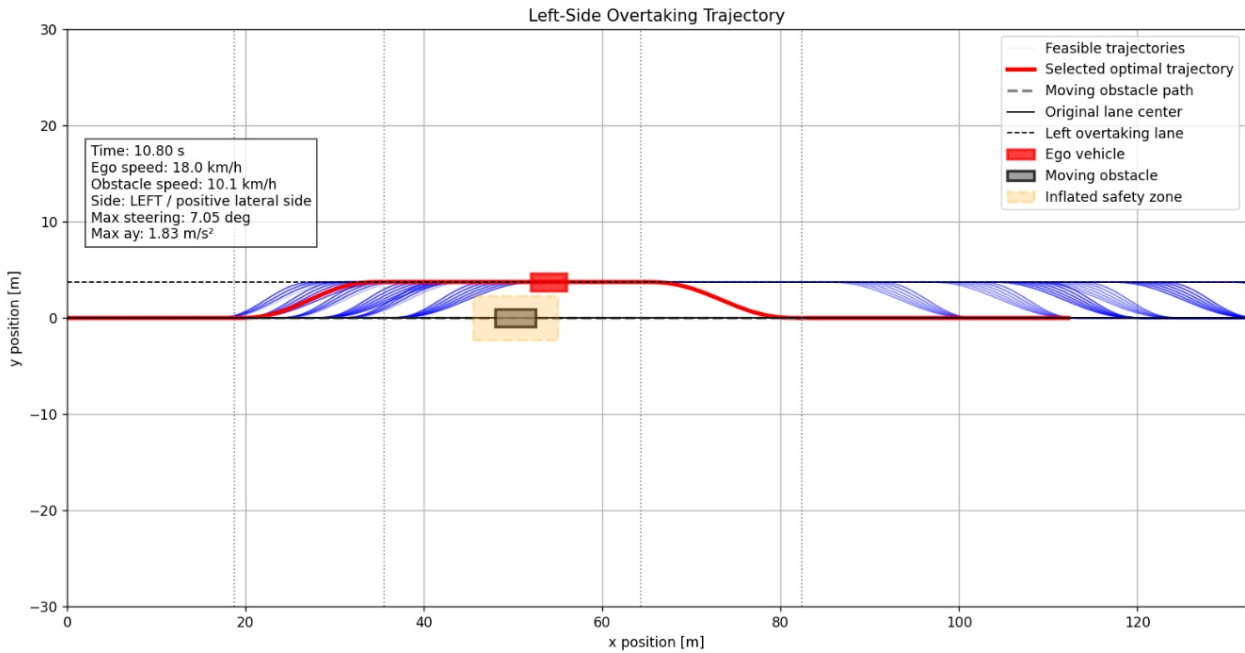


Table 9: Scenario 2 terminal output

Metric	Value	Limit / Target	Status
Ego speed	18.0 km/h (5.0 m/s)	4.0–5.0 m/s	✓
Feasible trajectories	2592	—	—
Obstacle speed	10.1 km/h (2.8 m/s)	2.8–3.8 m/s	✓
Ego speed	18.0 km/h (5.0 m/s)	—	—
Relative speed	2.2 m/s	≥ 1.0 m/s	✓
Outward lane-change length	16.8 m	—	—
Return lane-change length	18.0 m	—	—
Passing length	28.84 m	—	—
Cost value	15.2028	—	—
Max lateral acceleration	1.8329 m/s ²	≤ 3.8 m/s ²	✓ PASS
Max steering angle	7.047°	$\leq 25^\circ$	✓ PASS

Minimum clearance	1.95 m	≥ 0.5 m	✓ PASS
Outward main a_y	1.8329 m/s ²	—	—
Return main a_y	1.6045 m/s ²	—	—
Outward counter scale α	0.515	[0.40, 0.85]	✓
Return counter scale α	0.579	[0.40, 0.85]	✓
Decision mode	OVERTAKE	—	✓

The lateral acceleration profile (Figure 7) shows higher absolute values than Scenario 1 (peak 1.83 vs 1.09 m/s²) due to the shorter lane-change length (12–18 m vs 18–28 m) and the higher approach speed. Counter-steering pulses are smaller because the speed-penalty term in α is stronger with $v_{\text{rel}} = 2.2$ m/s. All values remain within the 3.8 m/s² limit with good margin.

Figure 7: Scenario 2 lateral acceleration profile

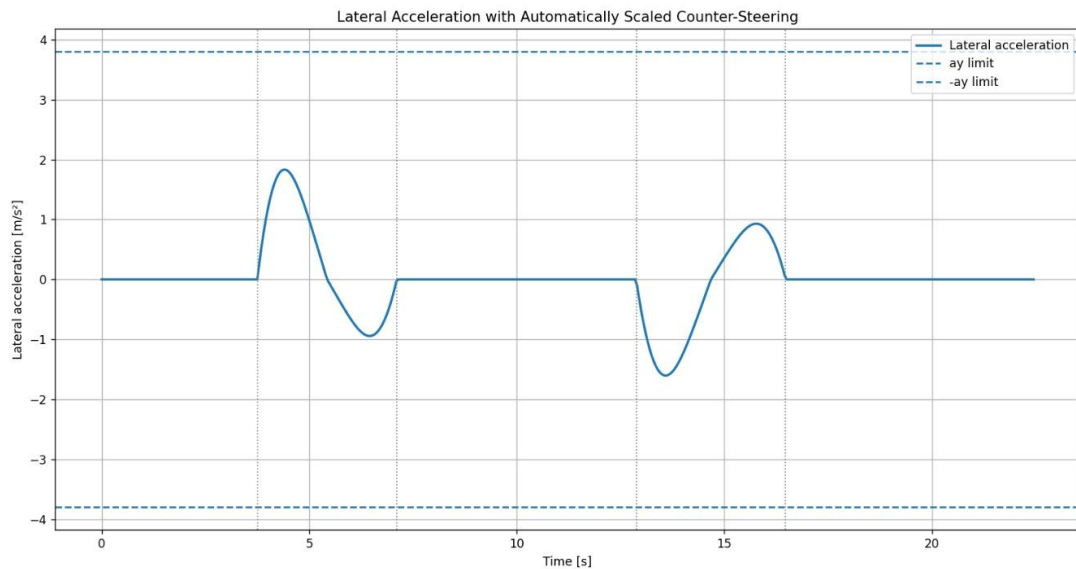
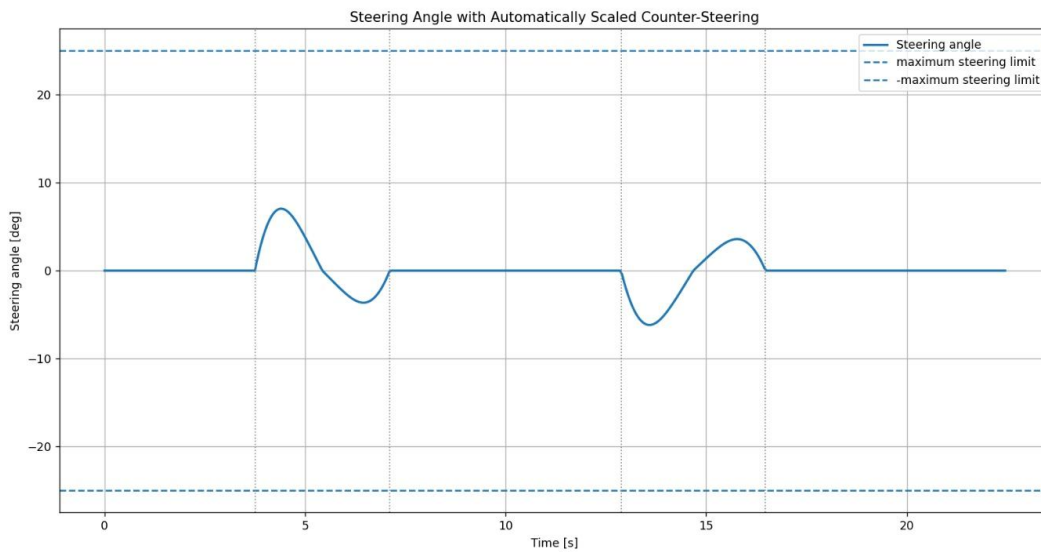


Figure 8: Scenario 2 steering angle profile

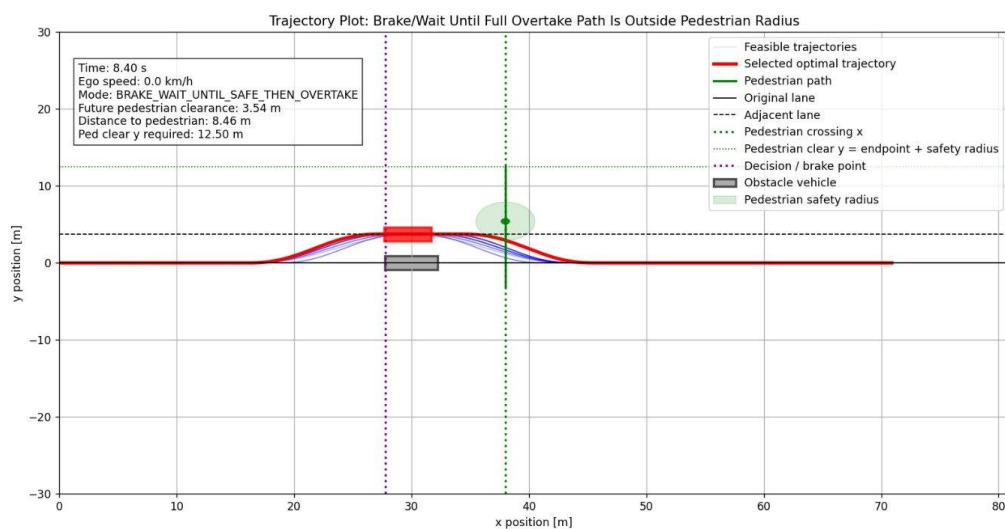


8.3 Scenario 3:

8.3.1 Overtaking with Pedestrian Crossing:

Scenario 3 is the most demanding scenario. The planner generated 64 candidate trajectories, of which 45 passed all geometric constraints. The rejection summary, selected trajectory parameters, counter-steering values, and safety results are shown in Figure 9.

Figure 9: Overtaking with Pedestrian Crossing



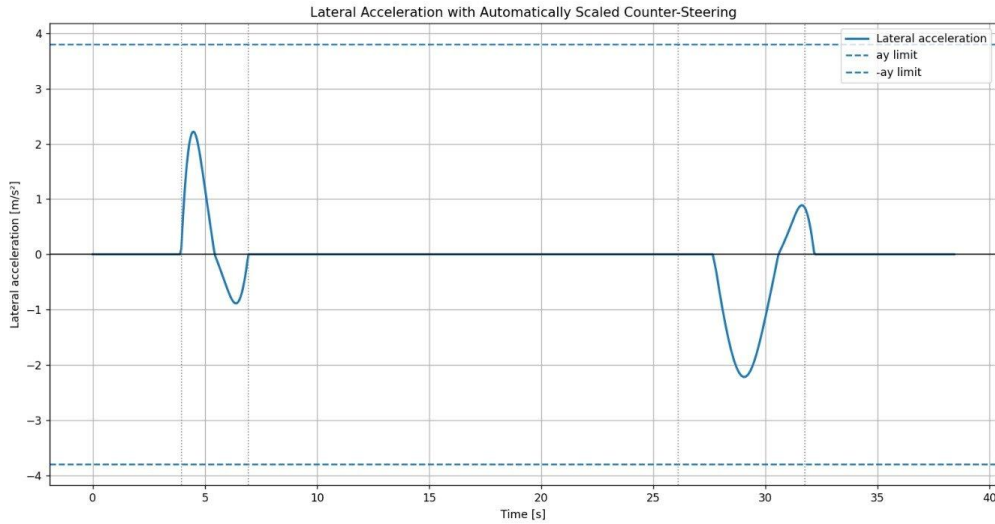
Key observation: the predicted ego-body margin at the original overtake decision point was -1.031 m (negative — the ego body would have entered the 2.75 m exclusion zone without intervention). The brake-wait-continue logic activated, braked the vehicle to rest, waited approximately 8 seconds for the pedestrian to clear $y = 12.5$ m, and resumed. The final pedestrian radius margin across the complete trajectory was $+3.555$ m — the exclusion zone was never violated.

Table 10: Scenario 3 terminal output

Metric	Value	Limit / Target	Status
Total trajectories	64	—	—
Feasible trajectories	45	—	—
Rejected bypass length	1,232	—	Most rejections
Rejected by lateral accel.	13	—	—
Rejected by obstacle clearance	6	—	—
Ego speed	4.0 m/s (14.4 km/h)	4.0–5.5 m/s	✓
Lane-change out length	12.000 m	—	—
Lane-change return length	12.000 m	—	—
Total maneuver length	28.500 m	—	—
Cost	21.859	Minimum	✓
Max lateral acceleration	2.221 m/s ²	≤ 3.8 m/s ²	✓ PASS
Max steering angle	13.171°	≤ 25°	✓ PASS
Obstacle clearance	1.779 m	≥ 0.5 m	✓ PASS
Future pedestrian clearance	3.543 m	≥ safety threshold	✓ PASS
Ped margin at decision	−1.031 m	≥ 0 → BRAKE	✗ → Brake activated
Final ped radius margin	3.555 m	≥ 0 m	✓ PASS
Outward main ay	2.2208 m/s ²	—	—
Outward counter ay	−0.8883 m/s ²	—	—
Return main ay	−2.2208 m/s ²	—	—
Return counter ay	0.8883 m/s ²	—	—
Outward counter scale α	0.400	[0.40, 0.85]	✓ (at clip)
Return counter scale α	0.400	[0.40, 0.85]	✓ (at clip)
Decision mode	BRAKE_WAIT_UNTIL_SAFE_THEN_OVERTAKE	—	✓

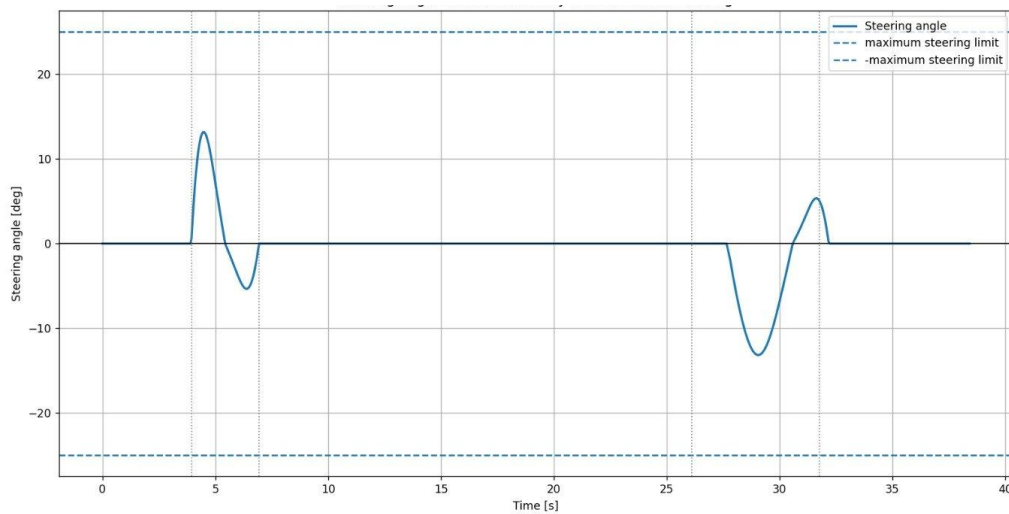
Figure 9 shows the Scenario 3 trajectory at $t = 8.40$ s: the ego (red) has stopped in the adjacent lane, waiting for the pedestrian (green dot with safety circle) who has not yet cleared PEDESTRIAN_CLEAR_Y = 12.5 m. The purple dashed line marks the decision/brake point; the green dotted line marks the required clearance y.

Figure 10: Scenario 2 lateral acceleration profile



The lateral acceleration profile (Figure 10) clearly shows the three-phase structure: outward pulse ($t \approx 4\text{--}7$ s), zero-acceleration wait ($t \approx 8\text{--}26$ s), and return pulse ($t \approx 28\text{--}32$ s) after the pedestrian clears. The steering angle (Figure 11) mirrors this structure, reaching a maximum of 13.17° — significantly higher than Scenario 2 (7.05°) because the ego speed is lower (4.0 m/s) with the same short lane-change length, producing higher curvature at that speed.

Figure 11: Steering angle profile

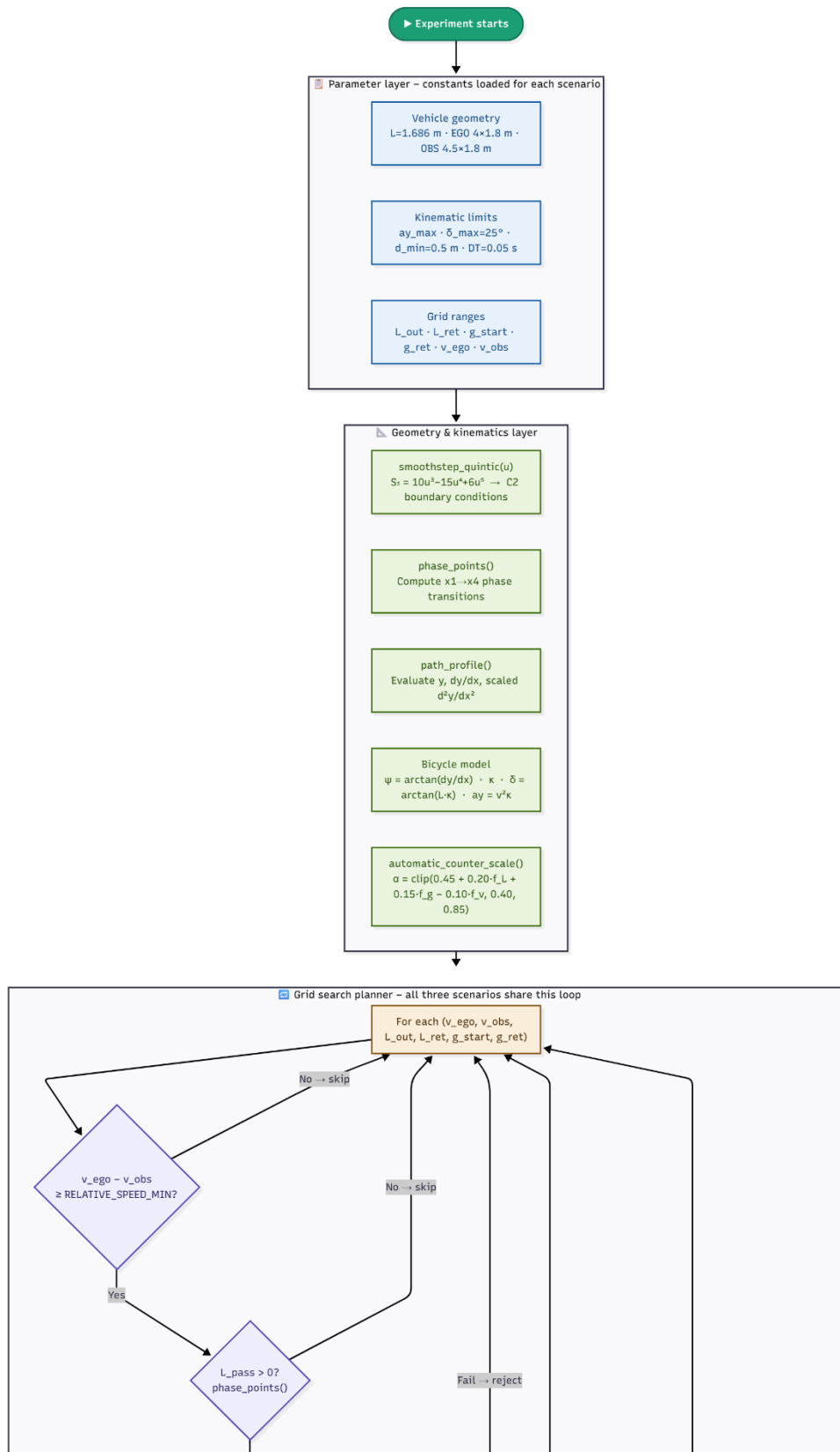


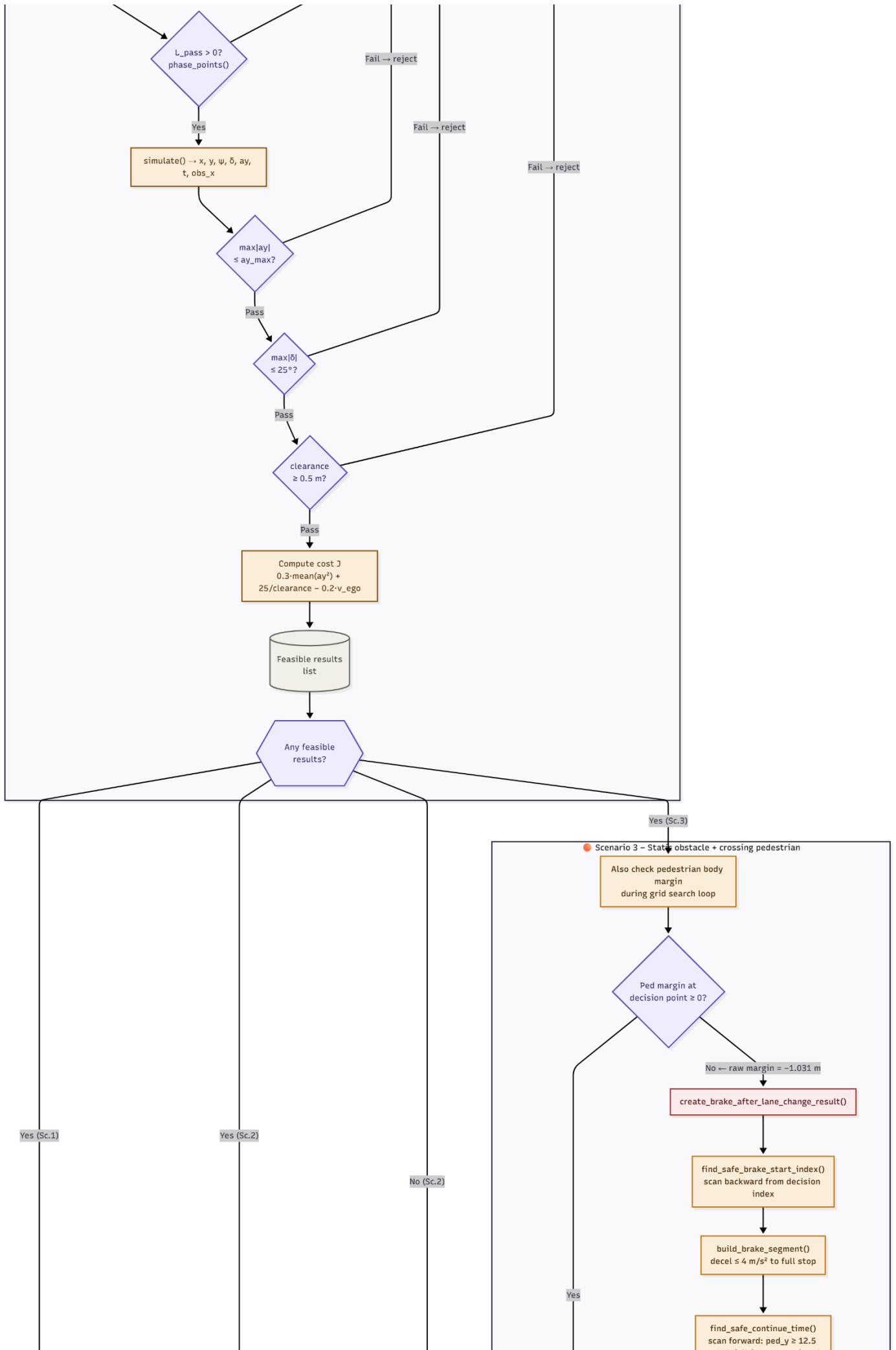
8.4 Cross-Scenario Comparison:

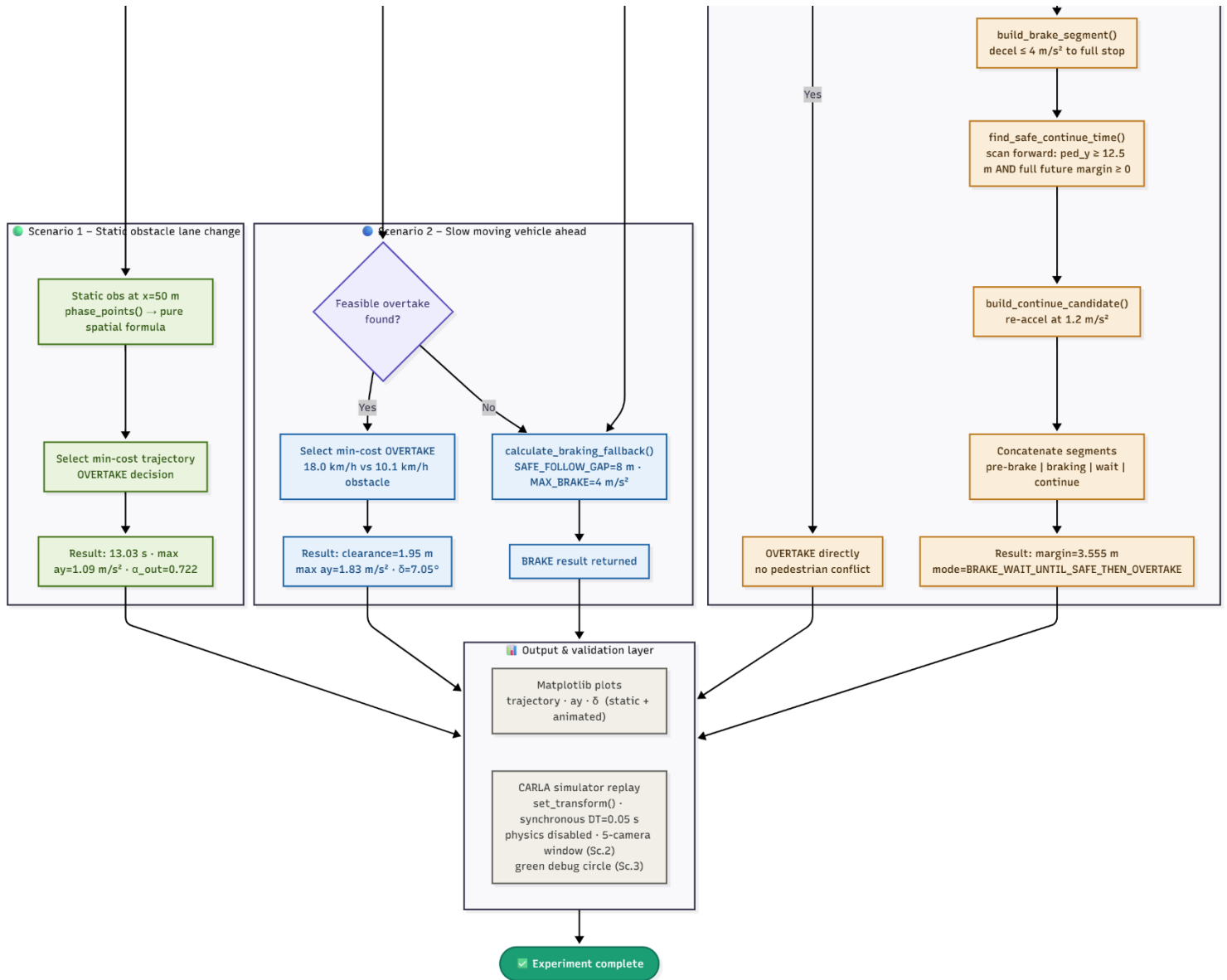
Table 11: Comparison between different scenarios

Metric	Scenario 1	Scenario 2	Scenario 3
Obstacle type	Static	Moving 10.1 km/h	Static + pedestrian
Ego speed	18 km/h	18.0 km/h	14.4 km/h
Max $ a_y $ [m/s ²]	1.09	1.83	2.22
Max $ \delta $ [deg]	~4.5	7.05	13.17
Min obstacle clearance	> 0.5 m	1.95 m	1.78 m
Pedestrian margin	—	—	3.555 m
Decision mode	OVERTAKE	OVERTAKE	BRAKE_WAIT_OVERTAKE
Total maneuver time	13.03 s	~22 s	~38 s
Feasible trajectories	2704	2592	45

9. Flow diagram:







10. Discussion:

10.1 Counter-Steering Effectiveness:

The adaptive counter-steering scaling produced physically meaningful results across all three scenarios. In Scenario 3, both α values hit the lower clip boundary (0.400) because the combination of the shortest available lane-change length (12.0 m), maximum start gap (10.0 m), and the lowest ego speed (4.0 m/s against a static obstacle) drive the formula below the physical minimum. The clip ensures a minimum counter-steering ratio is maintained regardless of extreme parameter combinations. Across Scenarios 1 and 2, α varied between 0.515 and 0.850, demonstrating the scaling mechanism working across its intended range.

10.2 Prospective vs. Reactive Pedestrian Safety:

The most significant architectural decision in this project was implementing a prospective pedestrian safety check rather than a purely reactive instantaneous-distance check. The -1.031 m raw margin at the Scenario 3 decision point illustrates why these matters: the ego vehicle was not yet close to the pedestrian at the moment of decision, but its intended future path would have brought the vehicle body into the exclusion zone during the passing phase. A reactive check operating only on current distance would have passed this situation through as safe, leading to a violation in a subsequent time step.

The iterative backward search for a safe braking index and the forward search for a safe resume time together ensure that the safety guarantee is prospective: the entire planned future trajectory is verified before any irreversible maneuver action is taken. This is directly analogous to the formal notion of safety in model predictive control, only commit to an action when the complete future plan is verified feasible, applied in a trajectory planning context.

10.3 Cost Function Sensitivity:

In Scenario 3, the top-5 ranked trajectories all selected $v_{\text{ego}} = 4.0$ m/s. This might seem counterintuitive given the speed reward term ($-0.2 \cdot v_{\text{ego}}$), but it reflects the combined effect of the other terms: at higher ego speeds against a static obstacle with short lane-change lengths, the $\text{mean}(a_y^2)$ term grows quickly (comfort penalty), and the maneuver length term also increases because the Doppler scaling formula produces larger x_4 values at higher speeds relative to a zero-speed obstacle (the denominator is always 1 in the static case, but the longer lane-change lengths needed to keep a_y within limits increase L_{maneuver}). The clearance barrier further suppresses candidates with marginal clearances. The cost function is working as designed.

10.4 CARLA Integration:

The synchronous-mode CARLA integration with physics disabled and direct `set_transform()` calls validates the planner's geometric outputs independently of any tracking controller. The five-camera Pygame window in Scenario 2 provided immediate qualitative confirmation of lane discipline and obstacle separation. The Scenario 3 debug circle confirmed that the ego body never visually entered the pedestrian safety zone during the replay. The minimum center-to-center distance observed in CARLA was consistent with the planned clearance metrics, validating that the local-to-CARLA coordinate transformation is correct.

11. Future Work:

- Integrate longitudinal speed profiling for adaptive acceleration during the overtaking phase.
- Replace constant-velocity prediction with intent-aware prediction (Gaussian process or LSTM).
- Extend to curved roads via Frenet-frame parameterization with clothoid reference curves.
- Interface with an MPC tracking controller (do-mpc / CasADi) for closed-loop CARLA validation.
- Implement multi-agent conflict resolution using social force models.
- Deploy on the ANTON platform (Renault Twizy Life) using the ROS 1 autonomous driving stack.

12. Conclusion:

This report has presented a trajectory planning system for automated vehicle evasive maneuvers across three scenarios of increasing complexity. The system uses quintic polynomial optimization-based trajectory planning (Optimal Control / MPC-like approach), path generation, kinematic bicycle model constraint evaluation, adaptive counter-steering, structured grid-search with cost-function ranking, and scenario-specific safety logic including a prospective pedestrian clearance check.

Section 3 demonstrated through a structured nine-criterion comparison that the quintic smoothstep approach outperforms RRT/RRT*, Hybrid A*, and trigonometric methods on C2 continuity, determinism, interpretability, and pedestrian interaction support, while matching Frenet polynomial planning on mathematical guarantees. The additional advantages over Frenet, no road geometry preprocessing, closed-form evaluation, natural pedestrian logic extension, make it the optimal choice for this straight-road scenario class.

The key technical contributions are: the adaptive counter-steering function, the Doppler-scaled phase point formulation, and the iterative prospective pedestrian safety verification. Experimental results confirmed that all kinematic constraints were satisfied across all scenarios, minimum clearances exceeded requirements, and Scenario 3 correctly triggered and resolved the pedestrian interaction with a final safety margin of 3.555 m. CARLA simulation replay visually validated all three scenarios under synchronous deterministic conditions.

13. References:

- [1] S. M. LaValle, "Rapidly-Exploring Random Trees: A New Tool for Path Planning," Tech. Rep. TR 98-11, Iowa State University, 1998.
- [2] S. Karaman and E. Frazzoli, "Sampling-based Algorithms for Optimal Motion Planning," *Int. J. Robot. Res.*, vol. 30, no. 7, pp. 846–894, 2011.
- [3] D. Dolgov, S. Thrun, M. Montemerlo, and J. Diebel, "Practical Search Techniques in Path Planning for Autonomous Driving," *Proc. AAAI Workshop AI in Robotics*, 2008.
- [4] Y. Ding et al., "Safe and optimal lane-change path planning for automated driving," *Proc. Inst. Mech. Eng. D*, vol. 235, no. 4, pp. 1070–1083, 2021. doi:10.1177/0954407020913735
- [5] R. Zhang et al., "Autonomous lane changing control for intelligent vehicles," *Cluster Comput.*, vol. 22(S4), pp. 8657–8667, 2019. doi:10.1007/s10586-018-1936-2
- [6] M. Werling, J. Ziegler, S. Kammel, S. Thrun, "Optimal Trajectory Generation for Dynamic Street Scenarios in a Frenet Frame," *Proc. IEEE ICRA*, pp. 987–993, 2010.
- [7] Y. Dou et al., "Lane changing prediction at highway lane drops using SVM and ANN," *IEEE AIM*, pp. 901–906, 2016. doi:10.1109/AIM.2016.7576883
- [8] D. Xie, "A Data-driven Lane-changing Model based on Deep Learning," *Transport. Res. Part C*, 2019. doi: 10.1016/j.trc.2019.07.002
- [9] R. Rajamani, *Vehicle Dynamics and Control*, 2nd ed. New York: Springer, 2012.
- [10] J. M. Maciejowski, *Predictive Control with Constraints*. Harlow: Prentice Hall, 2002.
- [11] B. Paden et al., "A Survey of Motion Planning and Control Techniques for Self-Driving Urban Vehicles," *IEEE Trans. Intell. Veh.*, vol. 1, no. 1, pp. 33–55, 2016.
- [12] J. D. Gammell et al., "Informed RRT*," *IEEE IROS*, pp. 2997–3004, 2014. doi:10.1109/IROS.2014.6942976
- [13] T. de Borba, "Project AI-001: Local Planning and Evaluation of Evasive Maneuvers – Project Brief," *CARISSMA C-ISAFE*, THI Ingolstadt, 2023.